

Education

Pontifical Catholic University of Peru (PUCP)

Mar 2019 – Dec 2024

BSc and Professional Title in Mechatronics Engineering

- Grade: 17.74/20 – **Rank: 1st** out of 80 students
- Thesis: Visual Navigation for wheeled quadruped with fall recovery capability (advised by [Prof. Diego Quiroz](#)).

University of Stuttgart

Oct 2023 – Mar 2024

Exchange Student in Computer Science

- Grade: 1.3 (scale 1–6, 1 best).
- Relevant Coursework: Foundation Models, Computer Vision, Deep Learning for NLP, Machine Learning.

Research Experience

Yale University

Remote, USA

Research Assistant (advised by [Prof. Josue Ortega](#))

Jan 2025 – May 2025

- Fine-tuned **Visual Language Models** (Llama Vision, Qwen-VL) on fMRI for clinical variable prediction.
- Developed a GPT-based predictor conditioned on a pretrained visual encoder for clinical variable prediction.
- Developed **Vector Quantized VAE** for fMRI reconstruction; aligned fMRI-text via **adversarial training**.

Max Planck Institute for Intelligent Systems (MPI-IS)

Stuttgart, Germany

Research Intern (advised by [Prof. Katherine Kuchenbecker](#))

Jul 2024 – Sep 2024

- Adapted Minsight, a vision based **soft tactile sensor**, for high frequency sensing by embedding a microphone.
- Developed **motion patterns** for a dexterous hand and a 3-axis machine to interact with fabrics and surfaces for data collection.
- Implemented **temporal models** to recognize textures dynamically in-hand using **multimodal** data.

Pontifical Catholic University of Peru (PUCP)

Lima, Peru

Research Assistant (advised by [Prof. Diego Arce](#))

Aug 2022 – Aug 2023

- **Compared traditional and DRL algorithms** for autonomous navigation of mobile robots in simulation.
- Tested mapping, localization, and evaluated traditional local planners in **highly dynamic environments**.
- Implemented and evaluated DRL algorithms for map-based and mapless autonomous navigation.

Industry Experience

Robot.com (formerly Kiwibot)

Hybrid, China-USA

Robotics Engineer

Aug 2025 – Present

- Developed **Isaac Sim digital twins** for robotic platforms, including tabletop systems and wheeled humanoids.
- Built software pipelines for humanoid **data collection** and VR/puppet-style **teleoperation**.
- Fine-tuned and evaluated **VLA models** including **Physical Intelligence** latest models via partnership.

Bosch Center for Artificial Intelligence (BCAI)

Renningen, Germany

Working Student

Oct 2023 – Mar 2024

- Developed hardware for robotic AI demonstrators: lane follower RC car and dice anomaly detector.
- Implemented an **Imitation Learning** pipeline for autonomous RC car and optimized inference with TensorRT.
- Automated data collection and trained **LSTM autoencoder** for audio-based anomaly detector demonstrator.

Publications

- **Solano J**, Quiroz D. “Stand, Walk, Navigate: Recovery-Aware Visual Navigation on a Low-Cost Wheeled Quadruped”. **IROS 2025 - Workshop on Wheeled-Legged Robotics**. [\[Paper\]](#)
- Andrussow I, **Solano J**, Richardson B, Martius G, Kuchenbecker K. “Adding internal audio sensing to internal vision enables human-like in-hand fabric recognition with soft robotic fingertips” (Oral). **Humanoids 2025**. [\[Website\]](#), [\[Paper\]](#)
- **Solano J**, Cisneros J, Sarmiento L, Quispe G, Hermitaño A, Quiroz D, Balbuena J. “Design and Implementation of an Inspection Robot for Crack Detection in Flooded Pipes.” **IEEE INTERCON 2023**. [\[Paper\]](#)
- Arce D, **Solano J**, Beltrán C. “A Comparison Study between Traditional and Deep-RL Algorithms for Indoor Autonomous Navigation in Dynamic Scenarios.” **Sensors 2023**. [\[Paper\]](#)

Competitions

European Rover Challenge

Remote, Poland

Computer Vision & Autonomous Navigation Member

2023, 2025

- (2023) Developed algorithms for **ArUco 3D pose estimation** and robot arm motion planning.
- (2025) Implemented 3D reconstruction and SLAM modules integrating depth and LiDAR data.

Latin American Space Challenge

São Paulo, Brazil

Electronics and Controls Lead

2021, 2023

- (2021) Designed and programmed the electronics system for a CanSat mini-satellite.
- (2023) Developed the **Go-to-goal PID navigation algorithm** for a comeback rover CanSat.

Teaching and Mentorship

Pontifical Catholic University of Peru (PUCP)

Oct 2025 - Present

Lecturer – ROS2 Specialization Program

Three courses (Introduction to ROS 2, Mobile Robot Navigation, and Robot Manipulation), focused on applied autonomy using ROS 2.

Mision Tech

Aug 2025

Mentor – Robotics Bootcamp

Mentored high school students through hands-on robotics and programming challenges.

Pontifical Catholic University of Peru (PUCP)

Mar 2025 – Mar 2026

Teaching Assistant – Robotics and AI (WS 25), Autonomous Vehicles (SS 26)

Assisted lectures and labs on kinematics, navigation, and AI, and helped test educational robotics modules.

Teens in AI

Mar 2022

Mentor – Women's Day Hackathon

Taught participants core machine learning concepts while mentoring teams building AI prototypes solutions.

Honors and Awards

First in Class

2025

Awarded distinction at the graduation ceremony for ranking 1st out of 80 Mechatronics graduates in the 2024 class.

CaCTüS Cohort

2024

9 selected from 700+ applicants worldwide to participate in a fully funded internship at the Max Planck Institute. [\[Link\]](#)

DARI Scholarship

2023

Awarded an exchange scholarship for academic excellence. Ranked 1st among PUCP exchange applicants. [\[Link\]](#)

Latin American Space Challenge

2023

Awarded 1st prize among 40+ teams from Latin America and Asia in the CanSat and Overall Satellite categories. [\[Link\]](#)

European Rover Challenge

2023

Finalists in the European Rover Challenge Remote 2023 [\[Link\]](#)

COAR National Scholarship

2018

Awarded full bachelor scholarship. Ranked 1st among 400+ applicants from high performing schools (COAR). [\[Link\]](#)

ONAM National Mathematics Olympiad

2017

Awarded 1st Place in the ONAM Trilce National Mathematics Olympiad. [\[Link\]](#)

Leadership and Community Service

PUCP Mechatronics Students Association

2023

Academic Affairs Representative: Supported academic community by organizing talks and compiling past exams.

PUCP Robotics Club

2022

Project Coordinator: Coordinated multidisciplinary robotics projects and organized technical workshops for competitions.

Skills

- **Programming:** Python, C/C++, MATLAB.
- **Tools & Simulation:** ROS 1/2, Gazebo, MuJoCo, Isaac Sim/Lab, Docker, Git, Google Cloud, SLURM.
- **Libraries & Frameworks:** Transformers, PyTorch, Tensorflow, OpenCV, MoveIt, Nav2.
- **Languages:** English (Fluent), Spanish (Native), German (Basic).